

Development of short test forms within item response theory: The shortIRT package

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Item Response Theory represents an ideal framework to reduce the length of existing tests thanks to the detailed information it provides on the measurement precision of each item with respect to different latent trait levels. shortIRT is an R package that implements recently introduced procedures for the development of short test forms (STFs). The item selection process of the newly introduced procedures is based on the definition of latent trait levels of interest, denoted as theta-targets, for which the information needs to be maximized. As such, the most informative item for each of the defined theta-targets is chosen to generate the STF. The shortIRT package allows the users to define the theta-targets on the latent trait using different strategies, to compare the results obtained from these strategies in terms of information, and to compute the difference between the true latent trait levels (or those estimated with the full-length test) and the latent trait estimates obtained from the STFs. The procedures implemented in the shortIRT package are conceptually similar to the automated test assembly procedures with maximin algorithms, where the aim is to maximize the information for a-priori defined levels of the latent trait. The main features of the shortIRT package are illustrated, together with some examples.